Supplementary material for

Exactness is not the limitation: a linear criterion for blind survivable sets on the parameter cube and complete classifications for four, five and six parameters

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Contents of the supplement

S1 proofs omitted from the main text (the slack game, the finite-horizon corollary in its sharp form, the feedback recursion); S2 the four-cube record and the pair tables for $m = 4, 5, 6$; S3 the five-cube record; S4 the six-cube record (clique orbits, maximal survivable orbits with representative cells and certificates, census, obstructions, boundary sets); S5 observation tables (probe ladders, feedback game, deadline horizons); S6 alpha-set tables and values at probed beliefs; S7 floating-point study and runtimes; S8 verification record (check list, Lean inventory, figure provenance); S9 symbols and conventions. Main-text theorem and equation numbers are quoted as "main Theorem " etc.; they are read from the compiled main text, and the verification script confirms the correspondence (check TXT.2). All tables are generated from `results.json` by `make_si_tables.py`.

S1 Proofs omitted from the main text

S1.1 Conventions

Cells $\theta \in \{\pm 1\}^m$, actions $u \in \{\pm 1\}^m \cup \{0\}$ (the hold is the zero vector), drift $D(u, \theta) = 2\langle u, \theta\rangle - 5$ in tenths, slack $s_\theta(t) = L + \sum_{\tau<t} D(u_\tau, \theta)$. A set S is kept by a word from L iff $s_\theta(t) \geq 0$ for all t and all $\theta \in S$.

S1.2 The slack game (main Proposition 4.2)

Proof. (a) If x is winning (for k steps) and $x' \geq x$ componentwise, the same word keeps x', since slacks only shift upward. The recursion for $\mathcal{W}_{k+1}$ is the definition; for the minimal elements note that $x \in \mathcal{W}_{k+1}$ iff $x + D(u) \geq w$ for some u and some $w \in \min \mathcal{W}_k$, i.e. $x \geq (w - D(u))^+$, because $x \geq 0$ anyway.

(b) $\mathcal{W} \subseteq \bigcap_k \mathcal{W}_k$ is trivial. Conversely, if $x \in \mathcal{W}_k$ for every k, the tree of words that keep x alive for their length is infinite and finitely branching ($|U_m| = 2^m + 1$), so it has an infinite branch by König's lemma, which is a word keeping x for ever.

(c) By definition of L^* and of blind survivability at level L (all cells start at the common slack L).

(d) Let $x \in \mathcal{W}^c$: a word keeps the clipped trajectory nonnegative for ever. The clipped trajectory is dominated by the unclipped one (clipping only lowers slacks, and the drifts do not depend on the state), so the same word keeps the unclipped trajectory nonnegative: $x \in \mathcal{W}$. Now let c dominate every coordinate of every minimal element of $\mathcal{W}$ and let $x \in \mathcal{W} \cap [0,c]^S$. Choose u with $x + D(u) \in \mathcal{W}$ (exists because $\mathcal{W}$ is the greatest fixed point of the one-step operator, by (b) and the recursion in (a)); then $x + D(u) \geq w$ for some minimal $w \in \mathcal{W}$, and $\min(x + D(u), c) \geq \min(w, c) = w$, so the clipped successor is again in $\mathcal{W} \cap [0,c]^S$. Repeating, the player stays in $\mathcal{W} \cap [0,c]^S$ for ever in the clipped game, i.e. $x \in \mathcal{W}^c$. The computation: $\mathcal{W}^c$ is the greatest fixed point of $X \mapsto \{x \in [0,c]^S : \exists u,\ \min(x + D(u), c) \in X\}$ on up-sets of the finite box, and the predecessor of an up-set with minimal elements M has minimal elements $\min\{(w - D(u))^+ : w \in M, u\} \cap [0,c]^S$: since $w_i \leq c$ for every minimal w of the box, $\min(x_i + D_i(u), c) \geq w_i$ iff $x_i + D_i(u) \geq w_i$, so the predecessor condition is $x \geq (w - D(u))^+$ exactly as in the unclipped game, intersected with the box. The iteration from $X_0 = [0,c]^S$ is decreasing in a finite lattice and stabilises.

(e) If a word keeps x_0 for t steps, its state after $\tau \leq t$ steps is dominated by an element of R_τ (induction: the successor of a dominated state is dominated by the successor of the dominating state, which is dominated by a maximal element of $R_{\tau+1}$). So $R_t = \emptyset$ means no word survives t steps, and the last nonempty R_{t-1} witnesses $t - 1$ steps. □

S1.3 The certified thresholds

For a set S the module computes $\mathcal{W}^c(S)$ for $c = 6, 12, 24, 48, 96$ in turn, records the least L with $(L, \ldots, L) \in \mathcal{W}^c(S)$, and then runs the forward reachability of (e) from $(L-1, \ldots, L-1)$ with a budget of 400 steps. If the frontier empties, $L^*(S) = L$ is certified; otherwise the next cap is tried. For every set of this paper the first cap at which the game won was already certified except for the subsets of the boundary ten-sets, where cap 6 returned 5 and cap 12 returned the certified 3. The number of sets with an uncertified threshold is 0 for $m = 4, 5, 6$ (Table S4.5).

S1.4 The sharp form of the finite-horizon corollary

Main Corollary 3.2 bounds the horizon by $\lfloor L/(5 - 2V) \rfloor$. The bound uses only the pooled inequality, so it is attained exactly when some word keeps the pooled slack $\sum_\theta q_\theta s_\theta(t)$ falling at the slowest rate $5 - 2V$ per step while no single cell of the support dies earlier. For a far pair at even m the constant action at distance $\frac{m-2}{2}$ from both cells gives both the drift $2m - 5 - 4 \cdot \frac{m-2}{2} = -1$, so each cell loses exactly one tenth per step and the horizon is L. For $m = 5$ and $h = 3$ every action is at distances (d_a, d_b) with $d_a + d_b \geq 3$, so the drifts are $(1, -3)$, $(-3, 1)$, $(-3, -3)$ or worse, and no action costs both cells exactly one tenth; the forward search gives $2\lfloor (L-1)/2 \rfloor$ for $L \leq 20$ (Table S2.2). For the whole cube ($V = 0$) the hold attains the bound exactly.

S1.5 The feedback recursion (main Proposition 6.3)

Let $F(T, L)$ be as defined in the main text. Soundness: if $L \geq L^*(T)$ a blind word keeps every cell of T; otherwise the splitting action u is played, the observed stock $L + D(u, \theta)$ identifies the class T_k of the true cell, and by induction (classes are strictly smaller) a strategy keeps every cell of every class from its slack. Completeness: consider any strategy that keeps every cell of T with the stock observed and let u be its first action. If u does not split T, the observation is uninformative and the strategy continues as a feedback strategy on T from slack $L + D(u, \cdot)$, which is constant on T; if that drift is nonnegative then $T \subseteq \mathrm{B}(u, r^*)$ is blind-survivable at the floor, and if it is negative the continuation shows $F(T, L')$ for some $L' < L$, hence $F(T, L)$ by monotonicity. If u splits T, the continuation is a feedback strategy on each class from its slack, so $F(T_k, L + k)$ holds for every class by induction, and $F(T, L)$ follows. The recursion is evaluated

with memoisation on (T, L) and the certified blind thresholds as the base case; the thresholds of the information classes met in the recursion at $m = 4, 5$ are all in $\{0, 1, 3, \infty\}$.

S2 The four-cube record and the pair tables

S2.1 Cube constants

Table S2.1 lists, for $m = 4, \ldots, 8$, the drift by number of mismatches, the radius $r^*(m)$ and size of the floor balls, the pair bound δ_m, Kleitman's bound $K(m)$ and the first-split cost $2m + 5$ (main Table 1 and Corollaries 3.3–3.5).

Table S2.1: Cube constants.

m	h=0	1	2	3	4	5	6	7	8	r^*	ball	δ_m	$K(m)$	$2m+5$
4	3	-1	-5	-9	-13					0	1	1	2	13
5	5	1	-3	-7	-11	-15				1	6	2	6	15
6	7	3	-1	-5	-9	-13	-17			1	7	3	12	17
7	9	5	1	-3	-7	-11	-15	-19		2	29	4	29	19
8	11	7	3	-1	-5	-9	-13	-17	-21	2	37	5	58	21

S2.2 Pairs

Table S2.2 gives, for every distance h and $m = 4, 5, 6$, the value $V = m - h$, survivability, the certified threshold of survivable pairs, and for the others the exact maximal survival horizon from $L = 0, 5, 10, 15, 20$ (forward reachability) against the LP bound $\lfloor L/(5-2V) \rfloor$ (main Corollary 3.2). The far pairs at distance $\delta_m + 1$ attain the bound at $m = 4, 6$ and fall short by one or two steps at $m = 5$ (S1.4).

Table S2.2: Pairs by distance: value, threshold, exact horizon versus LP bound.

m	h	V	survivable	L^*	exact horizon (LP bound) from $L = 0, 5, 10, 15, 20$
4	0	4	yes	0	–
4	1	3	yes	1	–
4	2	2	no	–	0: 0 (0); 5: 5 (5); 10: 10 (10); 15: 15 (15); 20: 20 (20)
4	3	1	no	–	0: 0 (0); 5: 1 (1); 10: 2 (3); 15: 4 (5); 20: 6 (6)
4	4	0	no	–	0: 0 (0); 5: 1 (1); 10: 2 (2); 15: 3 (3); 20: 4 (4)
5	0	5	yes	0	–
5	1	4	yes	0	–
5	2	3	yes	0	–
5	3	2	no	–	0: 0 (0); 5: 4 (5); 10: 8 (10); 15: 14 (15); 20: 18 (20)
5	4	1	no	–	0: 0 (0); 5: 1 (1); 10: 3 (3); 15: 5 (5); 20: 6 (6)
5	5	0	no	–	0: 0 (0); 5: 1 (1); 10: 2 (2); 15: 3 (3); 20: 4 (4)
6	0	6	yes	0	–
6	1	5	yes	0	–
6	2	4	yes	0	–
6	3	3	yes	1	–
6	4	2	no	–	0: 0 (0); 5: 5 (5); 10: 10 (10); 15: 15 (15); 20: 20 (20)
6	5	1	no	–	0: 0 (0); 5: 1 (1); 10: 2 (3); 15: 4 (5); 20: 6 (6)
6	6	0	no	–	0: 0 (0); 5: 1 (1); 10: 2 (2); 15: 3 (3); 20: 4 (4)

S2.3 The four-cube

G_4 is the four-cube graph; its 32 maximal cliques are the edges. The maximal survivable sets are the 32 adjacent pairs ($V = 3$, $L^* = 1$ certified: the saturated game at cap 6 has the antichain $\{(0,1),(1,0)\}$, and $(0,0)$ is lost in the first step). Table S2.3 is the certificate record; the census is in Table S4.4 together with the other cubes. The adjacent pair $\{a, b\}$ from $L = 1$ under the alternating word $a, b, a, b, \ldots$ has the slack trajectories $1, 4, 3, 6, 5, \ldots$ and $1, 0, 3, 2, 5, \ldots$; the matched action gives $+3$, the mismatched -1.

Table S2.3: Four-cube maximal survivable orbits with certificates.

size	orbit	V	$f(S)$	L^*	cap	antichain	lost after	word period / dip	boundary subsets
2	32	3	1	1	6	2	0	2 / 1	0

S3 The five-cube record

Table S3.4 lists the three orbits of maximal cliques of G_5 (distance ≤ 2) with the exact value and a maximiser; Table S3.5 the two orbits of maximal survivable sets with their certificates. The tetrahedron orbit (80 cliques of four cells pairwise at distance 2) has $V = 2$ with the uniform certificate: for the representative $\{(1,1,1,1,1),(1,1,1,-1,-1),(1,1,-1,1,-1),(1,1,-1,-1,1)\}$ the pooled cell is $(1,1,0,0,0)$. The square (a two-dimensional face) has $V = 3$ with the maximiser listed in Table S3.4, first-step bound 3 and certified threshold 3; the product word of the maximiser has period 4 and dip 3, and the opposite-corner alternation of the earlier editions is one of its orderings. Every clique of at most three cells of G_5 lies in a radius-one ball, so the census changes only at $L = 3$: 1,616 survivable sets (the nonempty subsets of the 32 balls, which overlap) for $L \leq 2$ and 1,696 for $L \geq 3$, against 1,776 pairwise-admissible sets.

Table S3.4: Maximal cliques of G_5 by orbit.

size	orbit	V	maximiser y	survivable
6	32	3	$(-1,1,-1,-1,1)$	yes
4	80	3	$(-1,1,0,0,1)$	yes
4	80	2	$(0,1,0,0,1)$	no

Table S3.5: Five-cube maximal survivable orbits with certificates.

size	orbit	V	$f(S)$	L^*	cap	antichain	lost after	word period / dip	boundary subsets
6	32	3	0	0	–	1	–	1 / 0	0
4	80	3	3	3	6	4	0	4 / 3	0

S4 The six-cube record

S4.1 Maximal cliques and their orbits

G_6 (distance ≤ 3) has 10,752 maximal cliques in seven orbits of B_6 ($|B_6| = 46{,}080$); Table S4.6 gives size, orbit size, exact value and a maximiser. The orbit decomposition is certified by comparing the orbit size of each representative (computed by counting stabilising group elements) with the size of its invariant class.

Table S4.6: Maximal cliques of G_6 by orbit (main Theorem 5.3).

size	orbit	V	maximiser y	survivable
12	192	3	$(-1,-1,0,1,1,-1)$	yes
10	1,280	$\frac{5}{2}$	$(-1,-1/2,1/2,1/2,1,-1)$	yes
8	160	3	$(0,-1,0,0,1,-1)$	yes
8	2,880	2	$(0,0,0,0,1,-1)$	no
8	3,840	2	$(0,0,0,0,1,-1)$	no
8	1,920	2	$(0,0,0,0,1,-1)$	no
8	480	2	$(0,0,0,0,1,-1)$	no

S4.2 Maximal survivable sets

Table S4.7 is the certificate record of the three maximal survivable orbits; the representatives, maximisers and the thresholds of all their nonempty subsets follow.

Table S4.7: Six-cube maximal survivable orbits with certificates; the last column counts the nonempty subsets of the representative with $V = \frac{5}{2}$ exactly.

size	orbit	V	$f(S)$	L^*	cap	antichain	lost after	word period / dip	boundary subsets
12	192	3	1	1	6	22	0	2 / 1	0
10	1,280	$\frac{5}{2}$	1	3	12	57	2	64 / 48	112
8	160	3	5	5	6	8	0	8 / 7	0

12 cells, orbit 192.
$y = (-1,-1,0,1,1,-1)$. Cells: `--+-`, `--+-`, `--+++-`, `--++++`, `-+-+-`, `-++-`, `-+++-`, `-++++`, `-+-++-`, `-+++++`, `+-++-`, `+-+++-`. Nonempty subsets by threshold: $L^* = 0$: 266, $L^* = 1$: 3,829.

10 cells, orbit 1,280.
$y = (-1,-1/2,1/2,1/2,1,-1)$. Cells: `--+-`, `--++-`, `-+-+-`, `-++-`, `-+++-`, `-++++`, `-+-++-`, `-++-+-`, `-+++++`, `+-+++-`. Nonempty subsets by threshold: $L^* = 0$: 161, $L^* = 1$: 750, $L^* = 3$: 112.

8 cells, orbit 160.
$y = (0,-1,0,0,1,-1)$. Cells: `--+-`, `--++-`, `-+-+-`, `-+++-`, `+--+-`, `+-++-`, `+-+-+-`, `+-+++-`. Nonempty subsets by threshold: $L^* = 0$: 72, $L^* = 1$: 182, $L^* = 5$: 1.

The maximal sets at each level (main Table 3) are: $L = 0$ the 64 radius-one balls; $L = 1, 2$ the 192 cylinders and 1,280 punctured subcubes (a three-dimensional subcube minus one vertex; $V = 3$ with $y = (-1,-1,-1,0,0,0)$ for the subcube $\theta_1 = \theta_2 = \theta_3 = -1$ minus $(-1,-1,-1,1,1,1)$; first-step bound and threshold 1; each lies in one full subcube and in one boundary ten-set); $L = 3, 4$ the cylinders and the boundary ten-sets; $L \geq 5$ all three maximal orbits.

S4.3 Census, thresholds, obstructions

Table S4.8: Level census for $m = 4, 5, 6$ (main Table 3 with all levels).

m	L	survivable	maximal	sizes	admissible	power set
4	0	16	16	1×16	48	2^{16}
4	1	48	32	2×32	48	2^{16}
5	0	1,616	32	6×32	1,776	2^{32}
5	1	1,616	32	6×32	1,776	2^{32}
5	2	1,616	32	6×32	1,776	2^{32}
5	3	1,696	112	4×80, 6×32	1,776	2^{32}
6	0	7,072	64	7×64	853,680	2^{64}
6	1	631,440	1,472	$7 \times 1{,}280$, 12×192	853,680	2^{64}
6	2	631,440	1,472	$7 \times 1{,}280$, 12×192	853,680	2^{64}
6	3	774,800	1,472	$10 \times 1{,}280$, 12×192	853,680	2^{64}
6	4	774,800	1,472	$10 \times 1{,}280$, 12×192	853,680	2^{64}
6	5	774,960	1,632	8×160, $10 \times 1{,}280$, 12×192	853,680	2^{64}

Table S4.9: Threshold histograms over all survivable sets, number of sets with $V = \frac{5}{2}$ exactly, and number of thresholds left uncertified by the two-sided scheme.

m	survivable sets	threshold histogram	$V = \frac{5}{2}$ exactly	uncertified
4	48	$L^* = 0$: 16, $L^* = 1$: 32	0	0
5	1,696	$L^* = 0$: 1,616, $L^* = 3$: 80	0	0
6	774,960	$L^* = 0$: 7,072, $L^* = 1$: 624,368, $L^* = 3$: 143,360, $L^* = 5$: 160	143,360	0

Table S4.10: Obstructions: far pairs by distance with their values, admissible non-survivable sets, minimal obstructions inside the admissible family by size and value, and the largest value of any minimal obstruction (main Corollary 5.4).

m	far pairs by distance	admissible non-survivable	minimal obstructions inside the admissible family	$\max V$
4	$h = 2$: 48 ($V = 2$), $h = 3$: 32 ($V = 1$), $h = 4$: 8 ($V = 0$)	0	none	2
5	$h = 3$: 160 ($V = 2$), $h = 4$: 80 ($V = 1$), $h = 5$: 16 ($V = 0$)	80	4 cells: 80 ($V = 2$)	2
6	$h = 4$: 480 ($V = 2$), $h = 5$: 192 ($V = 1$), $h = 6$: 32 ($V = 0$)	78,720	4 cells: 1,440 ($V = 2$), 5 cells: 960 ($V = 2$)	2

Representative minimal obstructions of the six-cube are the four-cell and five-cell sets

$$\{(1,1,-1,1,-1,1),\ (1,1,-1,1,1,-1),\ (1,1,1,-1,-1,-1),\ (1,1,1,-1,1,1)\},$$

$\{(1,1,-1,-1,1,1),\ (1,1,-1,1,-1,1),\ (1,1,-1,1,1,-1),\ (1,1,1,-1,-1,-1),\ (1,1,1,1,1,1)\}$,

both with $V = 2$; for the four-set the uniform certificate pools to $(1,1,0,0,0,0)$, for the five-set the certificate $q = (\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{3}, \frac{1}{6})$ (in the order listed) pools to $(1,1,0,0,0,0)$ as well.

S5 Observation tables

Table S5.11: Probe ladder for all levels at which it changes: blind value of the uniform belief on a j-dimensional subcube (main Table 4).

m	L	$j=6$	5	4	3	2	1	0
4	0			$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	1
4	1			$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	1	1
5	0	$\frac{3}{16}$	$\frac{5}{16}$	$\frac{1}{2}$	$\frac{3}{4}$	1	1	
5	1	$\frac{3}{16}$	$\frac{5}{16}$	$\frac{1}{2}$	$\frac{3}{4}$	1	1	
5	2	$\frac{3}{16}$	$\frac{5}{16}$	$\frac{1}{2}$	$\frac{3}{4}$	1	1	
5	3	$\frac{3}{16}$	$\frac{5}{16}$	$\frac{1}{2}$	1	1	1	
6	0	$\frac{7}{64}$	$\frac{3}{16}$	$\frac{5}{16}$	$\frac{1}{2}$	$\frac{3}{4}$	1	1
6	1	$\frac{3}{16}$	$\frac{5}{16}$	$\frac{1}{2}$	$\frac{7}{8}$	1	1	1
6	2	$\frac{3}{16}$	$\frac{5}{16}$	$\frac{1}{2}$	$\frac{7}{8}$	1	1	1
6	3	$\frac{3}{16}$	$\frac{5}{16}$	$\frac{1}{2}$	$\frac{7}{8}$	1	1	1
6	4	$\frac{3}{16}$	$\frac{5}{16}$	$\frac{1}{2}$	$\frac{7}{8}$	1	1	1
6	5	$\frac{3}{16}$	$\frac{5}{16}$	$\frac{1}{2}$	1	1	1	1

Table S5.12: Feedback game: blind and feedback thresholds of the whole cube, of subcubes and of the distance shells of a first action, $m = 4, 5$ (main Proposition 6.3).

m	class	cells	blind L^*	feedback F^*
4	whole cube	2^4	∞	23
4	subcube $j = 1$	2^1	1	1
4	subcube $j = 2$	2^2	∞	6
4	subcube $j = 3$	2^3	∞	14
4	shell $h = 0$	1	0	0
4	shell $h = 1$	4	∞	14
4	shell $h = 2$	6	∞	18
4	shell $h = 3$	4	∞	14
4	shell $h = 4$	1	0	0
5	whole cube	2^5	∞	18
5	subcube $j = 1$	2^1	0	0
5	subcube $j = 2$	2^2	3	3
5	subcube $j = 3$	2^3	∞	7
5	subcube $j = 4$	2^4	∞	11
5	shell $h = 0$	1	0	0
5	shell $h = 1$	5	0	0
5	shell $h = 2$	10	∞	11
5	shell $h = 3$	10	∞	11
5	shell $h = 4$	5	0	0
5	shell $h = 5$	1	0	0

Table S5.13: Total ignorance: exact maximal horizon of the whole cube from slack L, the LP bound and the number of holds affordable (main Proposition 6.4).

m	L	exact horizon	LP bound $\lfloor L/5 \rfloor$	hold steps
4	0	0	0	0
4	5	1	1	1
4	10	2	2	2
4	13	2	2	2
4	15	3	3	3
4	20	4	4	4
4	25	5	5	5
5	0	0	0	0
5	5	1	1	1
5	10	2	2	2
5	13	2	2	2
5	15	3	3	3
5	20	4	4	4
5	25	5	5	5
6	0	0	0	0
6	5	1	1	1
6	10	2	2	2
6	13	2	2	2
6	15	3	3	3
6	20	4	4	4
6	25	5	5	5

The cell-sum identity $\sum_\theta D(u,\theta) = -5 \cdot 2^m$ holds for all $2^m + 1$ actions at $m = 4, 5, 6$ (values $-80, -160, -320$), i.e. $-\frac{1}{2}$ per cell per period in stock units.

S6 Alpha-set tables and values at probed beliefs

Tables S6.14–S6.15 give $|\mathcal{A}_k(L)|$ for $k = 1, \ldots, 4$ and $L = 0, \ldots, 15$ at $m = 4$ (17^k words) and $m = 5$ (33^k words), with the number of masks that are blind-survivable sets in parentheses; the $k = 4$ columns reproduce the counts of the earlier scripts (`fivecube_alpha_check`). Tables S6.16–S6.17 give, for seven beliefs, the four finite-horizon values $\mathcal{V}_L^{(1)}/\mathcal{V}_L^{(2)}/\mathcal{V}_L^{(3)}/\mathcal{V}_L^{(4)}$ followed by the indefinite value $\mathcal{V}_L$. The beliefs: uniform; a far pair (mass $\frac{1}{2}$ on a cell and on a cell at distance $m-2$); an antipodal pair; a radius-one ball (mass $\frac{1}{2}$ spread uniformly) plus its antipode (mass $\frac{1}{2}$); three random beliefs with integer weights in $\{1, \ldots, 20\}$ drawn with `random.Random(7)` (the first three of the five computed). The equality $\mathcal{V}_L^{(4)} = \mathcal{V}_L$ holds for all beliefs at $L \le 3$ at $m = 4$ and at $L \le 4$ at $m = 5$ in these tables and fails at the next level, consistent with main Proposition 7.1: at $m = 5$ no non-survivable set outlives $2\lfloor (L-1)/2 \rfloor$ steps, which is below 4 for $L \le 4$.

Table S6.14: $|\mathcal{A}_k(L)|$ (survivable masks) at $m = 4$.

L	$k = 1$	$k = 2$	$k = 3$	$k = 4$
0	17 (16)	17 (16)	17 (16)	17 (16)
1	17 (0)	49 (48)	49 (48)	49 (48)
2	17 (0)	161 (48)	49 (48)	49 (48)
3	17 (0)	161 (48)	209 (48)	49 (48)
4	17 (0)	161 (48)	209 (48)	209 (48)
5	17 (0)	89 (48)	161 (48)	161 (48)
6	17 (0)	137 (0)	161 (48)	161 (48)
7	17 (0)	137 (0)	313 (48)	185 (48)
8	17 (0)	137 (0)	313 (48)	361 (48)
9	17 (0)	81 (0)	249 (48)	297 (48)
10	17 (0)	97 (0)	153 (48)	249 (48)
11	17 (0)	97 (0)	417 (0)	249 (48)
12	17 (0)	97 (0)	417 (0)	857 (48)
13	1 (0)	81 (0)	401 (0)	793 (48)
14	1 (0)	89 (0)	241 (0)	729 (48)
15	1 (0)	89 (0)	321 (0)	545 (48)

Table S6.15: $|\mathcal{A}_k(L)|$ (survivable masks) at $m = 5$.

L	$k = 1$	$k = 2$	$k = 3$	$k = 4$
0	33 (32)	465 (464)	625 (624)	945 (944)
1	33 (32)	465 (464)	945 (944)	945 (944)
2	33 (32)	705 (704)	945 (944)	945 (944)
3	33 (0)	545 (144)	1,185 (1,184)	1,345 (1,344)
4	33 (0)	513 (112)	1,185 (1,184)	1,505 (1,504)
5	33 (0)	513 (112)	3,185 (1,344)	5,265 (1,504)
6	33 (0)	945 (32)	3,185 (1,344)	5,265 (1,504)
7	33 (0)	425 (32)	2,345 (1,104)	3,505 (1,344)
8	33 (0)	393 (0)	2,025 (784)	6,065 (1,664)
9	33 (0)	393 (0)	5,385 (752)	9,625 (1,664)
10	33 (0)	705 (0)	4,105 (752)	7,705 (1,664)
11	33 (0)	275 (0)	2,867 (224)	6,937 (1,536)
12	33 (0)	307 (0)	2,707 (224)	14,649 (1,696)
13	33 (0)	307 (0)	4,595 (0)	15,929 (1,216)
14	33 (0)	265 (0)	2,035 (0)	12,937 (1,104)
15	1 (0)	233 (0)	2,195 (0)	11,145 (1,072)

Table S6.16: Values at probed beliefs, $m = 4$: $\mathcal{V}^{(1)}/\mathcal{V}^{(2)}/\mathcal{V}^{(3)}/\mathcal{V}^{(4)} \mid \mathcal{V}_L$.

L	uniform	far pair	antipodal pair	ball + antipode	random 1	random 2	random 3
0	$\frac{1}{16}/\frac{1}{16}/\frac{1}{16}/\frac{1}{16} \mid \frac{1}{16}$	$\frac{1}{2}/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{1}{2}/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{1}{2}/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{19}{146}/\frac{19}{146}/\frac{19}{146}/\frac{19}{146} \mid \frac{19}{146}$	$\frac{19}{161}/\frac{19}{161}/\frac{19}{161}/\frac{19}{161} \mid \frac{19}{161}$	$\frac{19}{178}/\frac{19}{178}/\frac{19}{178}/\frac{19}{178} \mid \frac{19}{178}$
1	$\frac{5}{16}/\frac{1}{8}/\frac{1}{8}/\frac{1}{8} \mid \frac{1}{8}$	$1/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{1}{2}/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{1}{2}/\frac{7}{10}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{35}{73}/\frac{18}{73}/\frac{18}{73}/\frac{18}{73} \mid \frac{18}{73}$	$\frac{67}{161}/\frac{38}{161}/\frac{38}{161}/\frac{38}{161} \mid \frac{38}{161}$	$\frac{75}{178}/\frac{37}{178}/\frac{37}{178}/\frac{37}{178} \mid \frac{37}{178}$
2	$\frac{5}{16}/\frac{5}{16}/\frac{1}{8}/\frac{1}{8} \mid \frac{1}{8}$	$1/1/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{1}{2}/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{1}{2}/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{35}{73}/\frac{35}{73}/\frac{18}{73}/\frac{18}{73} \mid \frac{18}{73}$	$\frac{67}{161}/\frac{67}{161}/\frac{38}{161}/\frac{38}{161} \mid \frac{38}{161}$	$\frac{75}{178}/\frac{75}{178}/\frac{37}{178}/\frac{37}{178} \mid \frac{37}{178}$
3	$\frac{5}{16}/\frac{5}{16}/\frac{5}{16}/\frac{1}{8} \mid \frac{1}{8}$	$1/1/1/\frac{1}{2} \mid \frac{1}{2}$	$\frac{1}{2}/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{1}{2}/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{35}{73}/\frac{35}{73}/\frac{35}{73}/\frac{18}{73} \mid \frac{18}{73}$	$\frac{67}{161}/\frac{67}{161}/\frac{67}{161}/\frac{38}{161} \mid \frac{38}{161}$	$\frac{75}{178}/\frac{75}{178}/\frac{75}{178}/\frac{37}{178} \mid \frac{37}{178}$
4	$\frac{5}{16}/\frac{5}{16}/\frac{5}{16}/\frac{5}{16} \mid \frac{1}{8}$	$1/1/1/1 \mid \frac{1}{2}$	$\frac{1}{2}/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{1}{2}/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{35}{73}/\frac{35}{73}/\frac{35}{73}/\frac{35}{73} \mid \frac{18}{73}$	$\frac{67}{161}/\frac{67}{161}/\frac{67}{161}/\frac{67}{161} \mid \frac{38}{161}$	$\frac{75}{178}/\frac{75}{178}/\frac{75}{178}/\frac{75}{178} \mid \frac{37}{178}$
5	$1/\frac{5}{16}/\frac{5}{16}/\frac{5}{16} \mid \frac{1}{8}$	$1/1/1/1 \mid \frac{1}{2}$	$1/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/\frac{35}{73}/\frac{35}{73}/\frac{35}{73} \mid \frac{18}{73}$	$1/\frac{67}{161}/\frac{67}{161}/\frac{67}{161} \mid \frac{38}{161}$	$1/\frac{75}{178}/\frac{75}{178}/\frac{75}{178} \mid \frac{37}{178}$
6	$1/\frac{1}{2}/\frac{5}{16}/\frac{5}{16} \mid \frac{1}{8}$	$1/1/1/1 \mid \frac{1}{2}$	$1/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/\frac{7}{10}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/\frac{45}{73}/\frac{35}{73}/\frac{35}{73} \mid \frac{18}{73}$	$1/\frac{96}{161}/\frac{67}{161}/\frac{67}{161} \mid \frac{38}{161}$	$1/\frac{50}{89}/\frac{75}{178}/\frac{75}{178} \mid \frac{37}{178}$
7	$1/\frac{1}{2}/\frac{5}{16}/\frac{5}{16} \mid \frac{1}{8}$	$1/1/1/1 \mid \frac{1}{2}$	$1/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/\frac{7}{10}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/\frac{45}{73}/\frac{35}{73}/\frac{35}{73} \mid \frac{18}{73}$	$1/\frac{96}{161}/\frac{67}{161}/\frac{67}{161} \mid \frac{38}{161}$	$1/\frac{59}{89}/\frac{75}{178}/\frac{75}{178} \mid \frac{37}{178}$
8	$1/\frac{1}{2}/\frac{5}{16}/\frac{5}{16} \mid \frac{1}{8}$	$1/1/1/1 \mid \frac{1}{2}$	$1/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/\frac{7}{10}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/\frac{45}{73}/\frac{35}{73}/\frac{35}{73} \mid \frac{18}{73}$	$1/\frac{96}{161}/\frac{67}{161}/\frac{67}{161} \mid \frac{38}{161}$	$1/\frac{59}{89}/\frac{75}{178}/\frac{75}{178} \mid \frac{37}{178}$
9	$1/\frac{1}{2}/\frac{5}{16}/\frac{5}{16} \mid \frac{1}{8}$	$1/1/1/1 \mid \frac{1}{2}$	$1/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/\frac{7}{10}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/\frac{45}{73}/\frac{35}{73}/\frac{35}{73} \mid \frac{18}{73}$	$1/\frac{96}{161}/\frac{67}{161}/\frac{67}{161} \mid \frac{38}{161}$	$1/\frac{59}{89}/\frac{75}{178}/\frac{75}{178} \mid \frac{37}{178}$
10	$1/1/\frac{5}{16}/\frac{5}{16} \mid \frac{1}{8}$	$1/1/1/1 \mid \frac{1}{2}$	$1/1/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/1/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/1/\frac{35}{73}/\frac{35}{73} \mid \frac{18}{73}$	$1/1/\frac{67}{161}/\frac{67}{161} \mid \frac{38}{161}$	$1/1/\frac{75}{178}/\frac{75}{178} \mid \frac{37}{178}$
11	$1/1/\frac{1}{2}/\frac{5}{16} \mid \frac{1}{8}$	$1/1/1/1 \mid \frac{1}{2}$	$1/1/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/1/\frac{7}{10}/\frac{1}{2} \mid \frac{1}{2}$	$1/1/\frac{45}{73}/\frac{35}{73} \mid \frac{18}{73}$	$1/1/\frac{96}{161}/\frac{67}{161} \mid \frac{38}{161}$	$1/1/\frac{59}{89}/\frac{75}{178} \mid \frac{37}{178}$
12	$1/1/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{8}$	$1/1/1/1 \mid \frac{1}{2}$	$1/1/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/1/\frac{7}{10}/\frac{7}{10} \mid \frac{1}{2}$	$1/1/\frac{45}{73}/\frac{45}{73} \mid \frac{18}{73}$	$1/1/\frac{96}{161}/\frac{96}{161} \mid \frac{38}{161}$	$1/1/\frac{59}{89}/\frac{59}{89} \mid \frac{37}{178}$
13	$1/1/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{8}$	$1/1/1/1 \mid \frac{1}{2}$	$1/1/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/1/\frac{7}{10}/\frac{7}{10} \mid \frac{1}{2}$	$1/1/\frac{45}{73}/\frac{45}{73} \mid \frac{18}{73}$	$1/1/\frac{96}{161}/\frac{96}{161} \mid \frac{38}{161}$	$1/1/\frac{59}{89}/\frac{59}{89} \mid \frac{37}{178}$
14	$1/1/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{8}$	$1/1/1/1 \mid \frac{1}{2}$	$1/1/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/1/\frac{7}{10}/\frac{7}{10} \mid \frac{1}{2}$	$1/1/\frac{45}{73}/\frac{45}{73} \mid \frac{18}{73}$	$1/1/\frac{96}{161}/\frac{96}{161} \mid \frac{38}{161}$	$1/1/\frac{59}{89}/\frac{59}{89} \mid \frac{37}{178}$
15	$1/1/1/\frac{1}{2} \mid \frac{1}{8}$	$1/1/1/1 \mid \frac{1}{2}$	$1/1/1/\frac{1}{2} \mid \frac{1}{2}$	$1/1/1/\frac{7}{10} \mid \frac{1}{2}$	$1/1/1/\frac{45}{73} \mid \frac{18}{73}$	$1/1/1/\frac{96}{161} \mid \frac{38}{161}$	$1/1/1/\frac{59}{89} \mid \frac{37}{178}$

Table S6.17: Values at probed beliefs, $m = 5$.

L	uniform	far pair	antipodal pair	ball + antipode	random 1	random 2	random 3
0	$\frac{3}{16}/\frac{3}{16}/\frac{3}{16}/\frac{3}{16} \mid \frac{3}{16}$	$\frac{1}{2}/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{1}{2}/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{1}{2}/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{73}{307}/\frac{73}{307}/\frac{73}{307}/\frac{73}{307} \mid \frac{73}{307}$	$\frac{97}{383}/\frac{97}{383}/\frac{97}{383}/\frac{97}{383} \mid \frac{97}{383}$	$\frac{3}{11}/\frac{3}{11}/\frac{3}{11}/\frac{3}{11} \mid \frac{3}{11}$
1	$\frac{3}{16}/\frac{3}{16}/\frac{3}{16}/\frac{3}{16} \mid \frac{3}{16}$	$\frac{1}{2}/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{1}{2}/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{1}{2}/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{73}{307}/\frac{73}{307}/\frac{73}{307}/\frac{73}{307} \mid \frac{73}{307}$	$\frac{97}{383}/\frac{97}{383}/\frac{97}{383}/\frac{97}{383} \mid \frac{97}{383}$	$\frac{3}{11}/\frac{3}{11}/\frac{3}{11}/\frac{3}{11} \mid \frac{3}{11}$
2	$\frac{3}{16}/\frac{3}{16}/\frac{3}{16}/\frac{3}{16} \mid \frac{3}{16}$	$\frac{1}{2}/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{1}{2}/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{1}{2}/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{73}{307}/\frac{73}{307}/\frac{73}{307}/\frac{73}{307} \mid \frac{73}{307}$	$\frac{97}{383}/\frac{97}{383}/\frac{97}{383}/\frac{97}{383} \mid \frac{97}{383}$	$\frac{3}{11}/\frac{3}{11}/\frac{3}{11}/\frac{3}{11} \mid \frac{3}{11}$
3	$\frac{1}{2}/\frac{5}{16}/\frac{3}{16}/\frac{3}{16} \mid \frac{3}{16}$	$1/1/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{1}{2}/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{3}{4}/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{190}{307}/\frac{116}{307}/\frac{73}{307}/\frac{73}{307} \mid \frac{73}{307}$	$\frac{237}{383}/\frac{159}{383}/\frac{97}{383}/\frac{97}{383} \mid \frac{97}{383}$	$\frac{113}{187}/\frac{81}{187}/\frac{3}{11}/\frac{3}{11} \mid \frac{3}{11}$
4	$\frac{1}{2}/\frac{5}{16}/\frac{3}{16}/\frac{3}{16} \mid \frac{3}{16}$	$1/1/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{1}{2}/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{3}{4}/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$\frac{190}{307}/\frac{116}{307}/\frac{73}{307}/\frac{73}{307} \mid \frac{73}{307}$	$\frac{237}{383}/\frac{159}{383}/\frac{97}{383}/\frac{97}{383} \mid \frac{97}{383}$	$\frac{113}{187}/\frac{81}{187}/\frac{3}{11}/\frac{3}{11} \mid \frac{3}{11}$
5	$1/\frac{5}{16}/\frac{5}{16}/\frac{5}{16} \mid \frac{3}{16}$	$1/1/1/1 \mid \frac{1}{2}$	$1/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/\frac{116}{307}/\frac{116}{307}/\frac{116}{307} \mid \frac{73}{307}$	$1/\frac{159}{383}/\frac{159}{383}/\frac{159}{383} \mid \frac{97}{383}$	$1/\frac{81}{187}/\frac{81}{187}/\frac{81}{187} \mid \frac{3}{11}$
6	$1/\frac{1}{2}/\frac{5}{16}/\frac{5}{16} \mid \frac{3}{16}$	$1/1/1/1 \mid \frac{1}{2}$	$1/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/\frac{3}{4}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/\frac{190}{307}/\frac{116}{307}/\frac{116}{307} \mid \frac{73}{307}$	$1/\frac{237}{383}/\frac{159}{383}/\frac{159}{383} \mid \frac{97}{383}$	$1/\frac{113}{187}/\frac{81}{187}/\frac{81}{187} \mid \frac{3}{11}$
7	$1/\frac{1}{2}/\frac{5}{16}/\frac{5}{16} \mid \frac{3}{16}$	$1/1/1/1 \mid \frac{1}{2}$	$1/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/\frac{3}{4}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/\frac{194}{307}/\frac{116}{307}/\frac{116}{307} \mid \frac{73}{307}$	$1/\frac{249}{383}/\frac{159}{383}/\frac{159}{383} \mid \frac{97}{383}$	$1/\frac{227}{374}/\frac{81}{187}/\frac{81}{187} \mid \frac{3}{11}$
8	$1/\frac{1}{2}/\frac{5}{16}/\frac{5}{16} \mid \frac{3}{16}$	$1/1/1/1 \mid \frac{1}{2}$	$1/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/\frac{3}{4}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/\frac{194}{307}/\frac{116}{307}/\frac{116}{307} \mid \frac{73}{307}$	$1/\frac{249}{383}/\frac{159}{383}/\frac{159}{383} \mid \frac{97}{383}$	$1/\frac{227}{374}/\frac{81}{187}/\frac{81}{187} \mid \frac{3}{11}$
9	$1/\frac{1}{2}/\frac{1}{2}/\frac{5}{16} \mid \frac{3}{16}$	$1/1/1/1 \mid \frac{1}{2}$	$1/\frac{1}{2}/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/\frac{3}{4}/\frac{3}{4}/\frac{1}{2} \mid \frac{1}{2}$	$1/\frac{194}{307}/\frac{190}{307}/\frac{116}{307} \mid \frac{73}{307}$	$1/\frac{249}{383}/\frac{237}{383}/\frac{159}{383} \mid \frac{97}{383}$	$1/\frac{227}{374}/\frac{113}{187}/\frac{81}{187} \mid \frac{3}{11}$
10	$1/1/\frac{1}{2}/\frac{5}{16} \mid \frac{3}{16}$	$1/1/1/1 \mid \frac{1}{2}$	$1/1/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/1/\frac{3}{4}/\frac{1}{2} \mid \frac{1}{2}$	$1/1/\frac{190}{307}/\frac{116}{307} \mid \frac{73}{307}$	$1/1/\frac{237}{383}/\frac{159}{383} \mid \frac{97}{383}$	$1/1/\frac{113}{187}/\frac{81}{187} \mid \frac{3}{11}$
11	$1/1/\frac{1}{2}/\frac{5}{16} \mid \frac{3}{16}$	$1/1/1/1 \mid \frac{1}{2}$	$1/1/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/1/\frac{3}{4}/\frac{1}{2} \mid \frac{1}{2}$	$1/1/\frac{194}{307}/\frac{116}{307} \mid \frac{73}{307}$	$1/1/\frac{249}{383}/\frac{159}{383} \mid \frac{97}{383}$	$1/1/\frac{227}{374}/\frac{81}{187} \mid \frac{3}{11}$
12	$1/1/\frac{1}{2}/\frac{1}{2} \mid \frac{3}{16}$	$1/1/1/1 \mid \frac{1}{2}$	$1/1/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/1/\frac{3}{4}/\frac{3}{4} \mid \frac{1}{2}$	$1/1/\frac{194}{307}/\frac{190}{307} \mid \frac{73}{307}$	$1/1/\frac{249}{383}/\frac{237}{383} \mid \frac{97}{383}$	$1/1/\frac{227}{374}/\frac{113}{187} \mid \frac{3}{11}$
13	$1/1/\frac{1}{2}/\frac{1}{2} \mid \frac{3}{16}$	$1/1/1/1 \mid \frac{1}{2}$	$1/1/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/1/\frac{5}{6}/\frac{3}{4} \mid \frac{1}{2}$	$1/1/\frac{201}{307}/\frac{194}{307} \mid \frac{73}{307}$	$1/1/\frac{253}{383}/\frac{249}{383} \mid \frac{97}{383}$	$1/1/\frac{237}{374}/\frac{227}{374} \mid \frac{3}{11}$
14	$1/1/\frac{1}{2}/\frac{1}{2} \mid \frac{3}{16}$	$1/1/1/1 \mid \frac{1}{2}$	$1/1/\frac{1}{2}/\frac{1}{2} \mid \frac{1}{2}$	$1/1/\frac{5}{6}/\frac{3}{4} \mid \frac{1}{2}$	$1/1/\frac{201}{307}/\frac{194}{307} \mid \frac{73}{307}$	$1/1/\frac{253}{383}/\frac{249}{383} \mid \frac{97}{383}$	$1/1/\frac{237}{374}/\frac{227}{374} \mid \frac{3}{11}$
15	$1/1/1/\frac{1}{2} \mid \frac{3}{16}$	$1/1/1/1 \mid \frac{1}{2}$	$1/1/1/\frac{1}{2} \mid \frac{1}{2}$	$1/1/1/\frac{3}{4} \mid \frac{1}{2}$	$1/1/1/\frac{194}{307} \mid \frac{73}{307}$	$1/1/1/\frac{249}{383} \mid \frac{97}{383}$	$1/1/1/\frac{227}{374} \mid \frac{3}{11}$

S7 Floating-point study and runtimes

The representative boundary ten-set (the first in the sorted record) has exact value $\frac{5}{2}$; SciPy's HiGHS returned 2.5 with error 0.0 for it. Its certified winning word from $L = 3$ (prefix of five actions, period four) is simulated in stock units ($z_0 = 1.3$, floor test $z \geq 1$) for 10^5 steps in `float64` and `float32`, with the drift coded as $-0.5 + 0.2\langle u, \theta\rangle$ ("affine") or as $(2\langle u, \theta\rangle - 5)/10$ ("ratio"); Table S7.18 reports false collapses, their first step and the maximal absolute deviation from the exact trajectory, together with the adjacent pair of the four-cube from $z_0 = 1.1$. Table S7.19 is the sweep over all 1,280 boundary ten-sets, each with its own certified word from $L = 3$, for 2,000 steps in `float64`. Table S7.20 gives the runtimes.

Table S7.18: Floating-point simulation of certified winning words.

instance	arithmetic / coding	false collapses	first at step	max. deviation
ten-set, $L = 3$	float64 affine	0	–	2.70e-08
ten-set, $L = 3$	float64 ratio	0	–	2.70e-08
ten-set, $L = 3$	float32 affine	3	3	1.15e+01
ten-set, $L = 3$	float32 ratio	3	3	1.15e+01
adjacent pair $m = 4$, $L = 1$	float64 affine	0	–	1.08e-08
adjacent pair $m = 4$, $L = 1$	float64 ratio	0	–	1.08e-08
adjacent pair $m = 4$, $L = 1$	float32 affine	0	–	6.13e+00
adjacent pair $m = 4$, $L = 1$	float32 ratio	0	–	6.13e+00

Table S7.19: Sweep over the 1,280 boundary ten-sets (2,000 steps, `float64`).

coding (float64)	sets with a false collapse	earliest step
$-0.5 + 0.2\langle u, \theta\rangle$	0	–
$(2\langle u, \theta\rangle - 5)/10$	1,120	3

Table S7.20: Runtimes (seconds, single core) of the full run of `run_all.py` and two sub-stages.

Stage	seconds
Classification $m = 4, 5, 6$ (cliques, orbits, LPs, certified thresholds, census)	147.1
Pair tables and horizons	0.1
Observation (ladders, feedback game $m = 4, 5$, deadline horizons)	10.1
Point-based sweeps and obstructions	443.9
Floating-point study (10^5-step simulations, 1,280-set sweep)	191.7
Total	792.9

Six-cube classification stage	seconds
Complete six-cube classification (inside the stage above)	146.9
Float sweep over the 1,280 boundary sets	175.8

S8 Verification record

S8.1 Check list

The verification script `paper05_exact_belief_computation_v20_verification.py` runs the checks of Table S8.21; the full mode recomputes the six-cube classification, the six-cube clique family by an independent algorithm, the five-cube alpha-sets and the floating-point sweep (about five minutes on one core), the `-quick` mode reuses the stored stage outputs for the expensive items. Every check is a boolean on exact rational or integer quantities; the final text checks parse the compiled main text and supplement and compare quoted numbers with `results.json`.

Table S8.21: Verification checks of the full run (identifier, statement, result). Items marked [full] are recomputed only in the full mode and read from the record in `-quick` mode.

id	check	result
A1.1	drift table: D = 2m-5-4h for m = 4..8 and all h; equals 10 d with d = -1/2 + <u,theta>/5	PASS
A1.2	floor balls: r*(m) = 0,1,1,2,2 and ball sizes 1,6,7,29,37 for m = 4..8	PASS
A1.3	cell-sum identity: sum over cells of D(u,theta) = -5*2^m for every action incl. the hold, m = 4,5,6	PASS
A1.4	the hold is the zero mixed action: D(hold) = -5 = 2<0,theta> - 5; first-split cost 2m+5 = cost of the antipode under a sign action	PASS
B1.1	exact LP: for every maximal-clique orbit representative (m = 4,5,6) the maximiser y and the certificate q attain the same value V, which equals the recorded V	PASS
B1.2	pairs: V({a,b}) = m - h and survivable iff h <= floor(m - 5/2) (= 1, 2, 3 for m = 4, 5, 6)	PASS
B1.3	product mixture of the maximiser has mean y and nonnegative expected drift on S; its periodic word (recorded period/dip) keeps every cell for three periods from the dip, for the three six-cube maximal orbits	PASS
B1.4	finite-horizon corollary on pairs: exact forward horizon <= floor(L/(5-2V)) for every non-survivable pair, L in {0,5,10,15,20}, m = 4,5,6, as recorded	PASS
B1.5	far pairs at distance delta_m + 1: horizon exactly L for L <= 20 at m = 4,6 (middle action costs -1 to both); 2*floor((L-1)/2) at m = 5	PASS
B1.6	deadline law: V(cube) = 0, the LP certificate and the uniform distribution both pool to the origin; exact horizon of the whole cube = floor(L/5) for L in {0,5,10,13,15,20,25}, m = 4,5,6	PASS

id	check (continued)	result
B1.7	tetrahedron: pairwise distance 2, every 3-subset inside a radius-one ball, V = 2 with pooled cell (1,1,0,0,0); exact horizon 2*floor((L-1)/2) for L <= 20 (as for the far pairs of the five-cube)	PASS
B1.8	Kleitman: K(m) = 2,6,12,29,58 for m = 4..8; extremal balls (odd m) are floor balls, extremal doubled balls (even m) have min <y,theta> = 3, first-step bound 1 and survive from slack 1 (certified for m = 4,6)	PASS
B1.9	support of the certificate q has at most m+1 cells for every maximal-clique representative	PASS
B1.10	floor census: the maximal sets at L = 0 are 2^m sets of the floor-ball size for m = 4,5,6 (Corollary on balls)	PASS
C1.1	adjacent pair (m = 4): antichain of minimal winning vectors {(0,1),(1,0)}, certified threshold 1, lost from 0 in the first step	PASS
C1.2	five-cube: square certified threshold 3 = first-step bound; ball threshold 0	PASS
C1.3	six-cube maximal orbits: certified thresholds 1 (cylinder), 3 (boundary ten-set), 5 (subcube) with the recorded caps and antichain sizes; first-step bounds 1, 1, 5	PASS
C1.4	cap artefact: the saturated game on the boundary ten-set returns 5 at cap 6 and 3 at cap 12; forward reachability from 2 empties within 2 steps	PASS
C1.5	boundary ten-set: extracted winning word from L = 3 has prefix 5 and period 4, keeps all cells for 50 periods, two zero-net cells touch the floor; pinned inequality sum q s(t) <= L holds for 200 steps	PASS
C1.6	boundary value: V(ten-set) = 5/2 exactly and every feasible maximiser is tight on supp q (checked for the LP maximiser)	PASS
C1.7	punctured subcube: V = 3, first-step bound 1, certified threshold 1; the full subcube has first-step bound 5 and threshold 5	PASS
D1.1	maximal cliques: 32 (m=4, all edges), 192 (m=5: 32 of size 6 + 160 of size 4), 10,752 (m=6: 192 x 12, 1,280 x 10, 9,280 x 8)	PASS
D1.2	orbit decomposition certified: invariant classes of maximal cliques (3 at m=5, 7 at m=6) each equal one orbit (orbit size of the representative = class size)	PASS
D1.3	maximal survivable sets recomputed from the clique representatives (certificate-support branching + global maximality): sizes {2}, {6,4}, {12,10,8}; orbit sizes 32; 32,80; 192,1280,160; totals 32, 112, 1,632	PASS
D1.4	six-cube census record: survivable sets 7,072 / 631,440 / 774,800 / 774,960 at L = 0 / 1-2 / 3-4 / >= 5; maximal 64 / 1,472 / 1,472 / 1,632 with the recorded sizes	PASS
D1.5	census consistency: threshold histogram sums to the number of survivable sets and its partial sums reproduce the census at every level; 143,360 = 1,280 x 112 boundary sets all at threshold 3	PASS
D1.6	four- and five-cube census: 16 / 48 (m=4, L = 0 / >= 1) and 1,616 / 1,696 (m=5, L <= 2 / >= 3); admissible families 48 and 1,776; maximal 16 singletons then 32 pairs; 32 balls then + 80 squares	PASS
D1.7	admissible families by an independent clique enumeration: 48 (m=4, also by direct subset enumeration) and 1,776 (m=5)	PASS
D1.8	[full] admissible family of the six-cube by independent enumeration: 853,680	PASS
D1.9	[full] complete six-cube classification recomputed: census, threshold histogram, knife-edge count and maximal orbits agree with the record	PASS
D1.10	minimal obstructions: none inside the admissible family at m=4, the 80 tetrahedra at m=5, 2,400 at m=6 (1,440 four-cell + 960 five-cell), all with V = 2; far pairs at distance delta_m+1 have V = 2; hence the uniform horizon L	PASS
D1.11	recorded six-cube obstruction examples: V = 2 and every proper subset survivable (minimality), pooled cell (1,1,0,0,0,0)	PASS
E1.1	probe ladder m=4: 1/16,1/8,1/4,1/2,1 at the floor and 1/8,1/4,1/2,1,1 from slack 1 (j = 4..0)	PASS
E1.2	probe ladder m=5: 3/16,5/16,1/2,3/4,1,1 for L <= 2 and 3/16,5/16,1/2,1,1,1 for L >= 3; m=6: 7/64,...,3/4,1,1 at the floor, 3/16,5/16,1/2,7/8,1,1,1 for 1 <= L <= 4, 3/16,5/16,1/2,1,1,1,1 for L >= 5	PASS
E1.3	ladders m = 4,5 recomputed from the stored maximal families census_m4/5.npz (max over subcubes of the mass of a maximal set) agree with the record	PASS
E1.4	feedback game recomputed: F*(cube) = 23 (m=4) and 18 (m=5); subcubes 1,6,14 and 0,3,7,11; shells 0,14,18,14,0 and 0,0,11,11,0,0	PASS
E1.5	first-split decomposition: 23 = max(13, 9+14, 5+18, 1+14) at m=4 and 18 = max(15, 3+11, 7+11, 11) at m=5; the shells at m=5 with h in {1,4} are blind-survivable (threshold 0)	PASS
E1.6	blind thresholds of the whole cube and of antipodal sets are infinite (V < 5/2); subcube blind thresholds: m=4: 1, inf, inf; m=5: 0, 3, inf, inf	PASS
F1.1	alpha-set sizes m=4, k=4, L = 0..15 recomputed over 17^4 words: 17,49,49,49,209,161,161,185,361,297,249,249,857,793,729,545	PASS
F1.2	[full] alpha-set sizes m=5, k=4 recomputed over 33^4 words equal the record and the earlier fivecube log	PASS
F1.3	values at probed beliefs: V^(k) nonincreasing in k and >= indefinite value; equal at k = 4 for L <= 3 (m=4) and L <= 4 (m=5) and not at the next level; far-pair belief V^(4) = 1 for L >= 4 and indefinite 1/2; uniform indefinite 1/8 (m=4, L >= 1)	PASS

id	check (continued)	result
F1.4	m=4, L=15: survivable masks among the k-step alpha-sets are 0, 0, 0, 48 for k = 1..4; uniform V^(4) = 1/2 vs indefinite 1/8; m=5, L=15: uniform 1/2 vs 3/16	PASS
F1.5	definition-level value at the uniform belief (m=4, L=15, k=2) by direct enumeration of 17^2 words equals the alpha-set value (1)	PASS
F1.6	pairs of the four-cube by distance: 32 adjacent, 48 at 2, 32 at 3, 8 antipodal (120 in all)	PASS
G1.1	arithmetic of the pinned cell: three drifts of -1 tenth from z = 1.3 give exactly the floor in exact arithmetic; float32 and float64 with (2k-5)/10 land below 1, float64 with -0.5+0.2k lands at or above 1	PASS
G1.2	recorded simulations: float32 false collapse at step 3 for the representative ten-set; float64 none in 10^5 steps (both codings); float LP value 2.5 for V = 5/2	PASS
G1.3	[full] float64 sweep over the 1,280 boundary ten-sets recomputed: ratio coding 1,120 false collapses (earliest step 3), affine coding 0	PASS
G1.4	runtimes recorded for every stage; total within 2 s of the sum of the stages	PASS
L1.1	Lean layer: 8 modules, 80 theorem declarations (Dynamics 28, Deadline 19, Pairs 11, Hamming 6, Classification 5, ExactBelief 5, Bands 4, Ladder 2), no sorry, no axiom declarations; the ten named footprints present	PASS
TXT.1	compiled numbering of the results quoted in captions and the supplement: thm:criterion = 3.1, cor:horizon = 3.2, cor:pairs = 3.3, cor:balls = 3.4, cor:kleitman = 3.5, prop:incomparable = 3.6, prop:game = 4.2, prop:firststep = 4.4, prop:pinned = 4.5, thm:m4 = 5.1, thm:m5 = 5.2, thm:m6 = 5.3, cor:uniform = 5.4, prop:antichain = 6.1, prop:feedback = 6.3, prop:deadline = 6.4, prop:pbvi = 7.1, fig:census = 1, fig:horizon = 2, fig:knife = 3	PASS
TXT.2	supplement's main-text number table agrees with the compiled .aux on every label	PASS
TXT.3	headline tokens present in the assembled main text: 774{,}960, 853{,}680, 143{,}360, 1{,}120, 10{,}752, 631{,}440, 774{,}800, 7{,}072, 1{,}632, 2{,}400, 1{,}616, 1{,}696, 1{,}776, \{}$Fstar$(\{}$Theta_4$) = 23, \{}$Fstar$(\{}$Theta_5$) = 18, $5T$, 857, 15{,}929, 5{,}373, 80 theorem, K(8)=58	PASS
TXT.4	the number of checks quoted in the manuscript (\{}NCHECKS) equals the number of checks of this script	PASS

S8.2 Lean inventory

Table S8.22: Lean modules: theorem declarations (including `@[simp]`-attributed ones) and `sorry` count; the modules declare no axioms.

module	theorem declarations	`sorry`
`EBC_Bands_v2.lean`	4	0
`EBC_Classification_v3.lean`	5	0
`EBC_Deadline.lean`	19	0
`EBC_Dynamics.lean`	28	0
`EBC_ExactBelief_v3.lean`	5	0
`EBC_Hamming.lean`	6	0
`EBC_Ladder_v2.lean`	2	0
`EBC_Pairs_v3.lean`	11	0
total	80	0

Declared theorems by module. **EBC_Bands_v2.lean**: `matched_drift_scale_m4`, `matched_drift_pos_m4`, `drift_matched_pos`, `singleton_survives`.

EBC_Classification_v3.lean: `five_add_five_eq_ten`, `four_sub_five_eq_neg_one`, `one_sub_eq_neg_sub`, `far_pair_sum_bound_expanded`, `far_pair_horizon_bound`.

EBC_Deadline.lean: `holdRep_zero`, `matchedRep_zero`, `holdRep_length`, `matchedRep_length`, `holdAct_drift`, `matched_drift_nonneg`, `deadline_half_pos`, `deadline_half_nonneg`, `totalDrift_holdRep`, `deadline_necessity`, `matchedRep_survives`, `holdRep_survives`, `survivesTo_append`, `deadline_sufficiency`, `deadline_law`, `holdAct_isFullHold`, `matchedRep_mem_inAlphabet`, `deadlineSeq_mem_inAlphabet`, `deadlineSeq_inAlphabet`.

EBC_Dynamics.lean: `natToK_zero`, `natToK_succ`, `natToK_nonneg`, `natToK_add`, `natToK_mono`, `lsum_ones`, `natToK_le_one`, `neg_add`, `two_pos`, `two_ne_zero`, `five_pos`, `five_ne_zero`, `half_mul_two`, `half_add_half`,

five_mul_fifth, stateAfter_eq, survivesTo_final, survivesTo_totalDrift_lower, lsum_neg_ones, drift_pair, totalDrift_pair, lsum_const, pairIpSum_upper, pairIpSum_lower_of_survive, le_of_mul_le_mul_pos, agreeMass_add_natToK_hamming, cor_hamming, cor_hamming_dist.

EBC_ExactBelief_v3.lean: hamming_eq_zero, hamming_ne_zero_exists, hamming_zero_of_agree, no_adjacency_triangle, no_three_pairwise_adjacent.

EBC_Hamming.lean: natToK_mul, two_mul_two_eq_four, two_mul_four_eq_five_add_three, le_of_add_le_add_left, three_lt_four, cor_hamming_m4.

EBC_Ladder_v2.lean: no_three_distinct_above, survivor_is_singleton_or_adjacent_pair.

EBC_Pairs_v3.lean: add_lt_add_right', eq_of_mul_eq_mul_pos, fifth_pos, tenth_le_half, half_fifth_two_eq_neg_tenth, ipi_cell_le_two, drift_cell_le_neg_tenth, step_fails_cell, ipi_hold_zero, step_fails_hold, pair_first_step_fails.

S8.3 Figure provenance

Figure 1 (census by level) is drawn from the `classification` record; Figure 3 (knife-edge word and floating-point deviations) recomputes the certified word of the first two boundary ten-sets of the sorted record with the exact module and simulates the pinned cell in `float64` with both codings; Figure 2 (horizons and the feedback price) is drawn from the `pairs` and `observation` records. All three are produced by `make_figs.py`.

S9 Symbols and conventions

m; $\mathbf{\Theta}_m = \{\pm 1\}^m$	number of uncertain signs; cells
$U_m = \{\pm 1\}^m \cup \{\circ\}$	actions (sign vectors and the hold, identified with $0 \in \mathbb{R}^m$)
d_{H}	Hamming distance; h = number of mismatched signs
$d(u,\theta) = -\frac{1}{2} + \frac{1}{5}\langle u, \theta\rangle$	drift in stock units; floor $z \geq 1$
$D(u,\theta) = 2\langle u,\theta\rangle - 5 = 2m - 5 - 4h$	drift in tenths; $D(\circ,\theta) = -5$
$L = 10(z_0 - 1)$; $s_\theta(t)$	initial slack in tenths; slack of cell θ at time t
$\mathcal{S}_m(L)$; $\mathcal{M}_m(L)$	blind-survivable sets at level L; their maximal members
$L^*(S)$; $f(S)$	threshold; first-step bound $\min_u \max_\theta (-D)^+$
$V(S) = \max_y \min_\theta \langle \theta, y\rangle$	value of the linear criterion; survivable iff $V \geq \frac{5}{2}$
$q \in \Delta(S)$; $\sum_\theta q_\theta \theta$	certificate of non-survivability; pooled cell
$r^*(m) = \lfloor (2m-5)/4 \rfloor$	radius of the floor balls
$\delta_m = \lfloor m - \frac{5}{2} \rfloor$; $K(m)$	pair bound on the distance; Kleitman's bound
G_m	graph on cells with edges at distance $\leq \delta_m$ (admissible pairs)
B_m	hyperoctahedral group, $\lvert B_m\rvert = 2^m m!$
$\mathcal{W}(S)$, $\mathcal{W}_k(S)$, $\mathcal{W}^c(S)$	winning regions: indefinite, k-step, saturated at cap c
$\mathcal{V}_L(b)$; $\mathcal{V}_L^{(k)}(b)$	blind value of a belief; k-step value
$\mathcal{A}_k(L)$	alpha-set of k-step outcome masks
$F(T,L)$; $F^*(T)$	feedback survivability; feedback threshold

Sign conventions: slacks and drifts are integers in tenths; "survives" means $s_\theta(t) \geq 0$ for all t; a "false collapse" is a floating-point stock below 1 at a time when the exact stock is ≥ 1. The cells of a representative are written as sign strings (`+` for $+1$, `-` for -1).